

Land Type Classification

Using Sentinel-2 Satellite Images & Deep Learning

EuroSAT Dataset

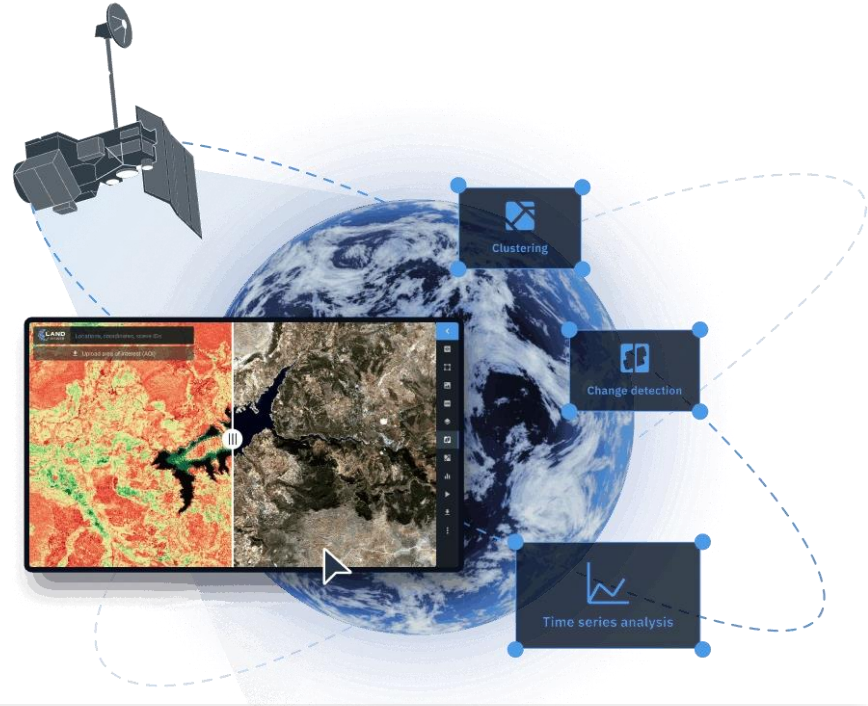
CNN / Transfer Learning

10 Land Classes

PyTorch

Project Overview

- Our project involves classifying Land use land cover (LULC) using satellite imagery captured by the Sentinel-2 optical sensor.
- Sentinel-2 is a European satellite that images the Earth every five days with a spatial resolution of up to 10 meters.
- The project focuses on classifying land cover into types or categories using deep learning algorithms



The Problem

Challenge

Satellite data is massive and complex, manually labeling land use is slow, expensive, and unscalable.

Accurate land classification is critical for:

- Urban & infrastructure planning
- Agriculture & crop monitoring
- Environmental conservation
- Resource management

Our Objective

Build a Deep Neural Network that automatically classifies land types from Sentinel-2 multispectral satellite images, across 10 categories, with high accuracy.

Dataset: EuroSAT (Sentinel-2)



27,000

Labeled Images

10 Meters

Spatial Resolution

64×64

Pixels / Image

13 Bands

Multi-Spectral Data

10 Land Classes:

→ Used RGB bands (2,3,4) for generalizability

1. Annual Crop

3. Herbaceous Veg.

5. Industrial

7. Permanent Crop

9. River

2. Forest

4. Highway

6. Pasture

8. Residential

10. Sea / Lake

Project Pipeline

1

Data Collection & Processing

Download EuroSAT 27K images
Convert .tif to JPEG RGB bands
80/10/10 train/val/test split

2

EDA & Preprocessing

Class distribution analysis
Spectral band visualization
Augmentation: flip, rotate, crop

3

Model Development

ResNet50 Transfer Learning
RGB model + MS model
Custom FC head + Dropout

4

Evaluation & Tuning

Accuracy, F1-score
Confusion matrix
Train/val loss curves

5

Deployment

Streamlit Cloud deployment
Satellite image upload & classify
Sliding-window segmentation map

Data Processing



NORMALIZATION

Scaling pixel values to $[0, 1]$ for RGB and Sentinel-standard for TIF.



AUGMENTATION

Random flips, rotations, and brightness shifts to boost robustness.



DATA SPLIT

70% Train, 20% Val, 10% Test using stratified sampling.



OPTIMIZATION

PyTorch pipelines with caching and prefetching.



Reproducibility: Fixed random seeds used for consistent dataset shuffling and splitting.

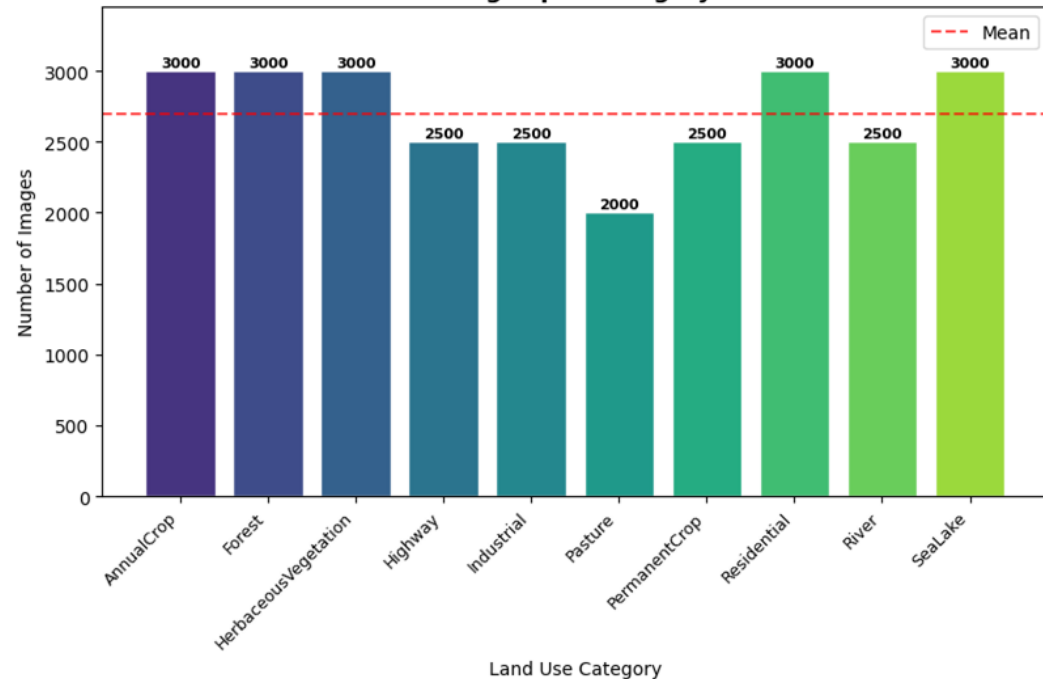


Efficiency: Parallelized loading and RAM caching to minimize training bottlenecks.

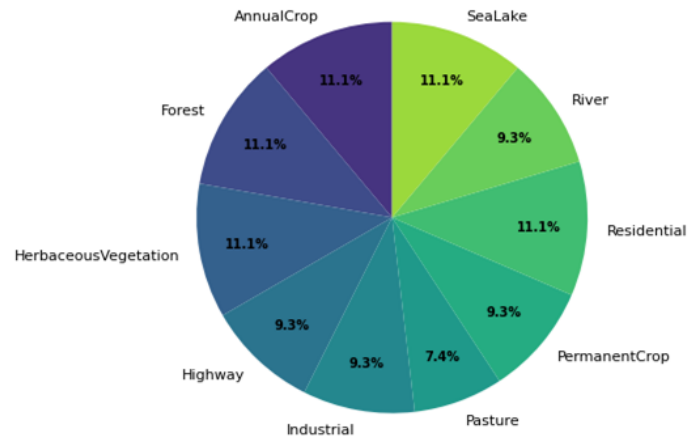
Exploratory Data Analysis Results

Class Distribution Charts

Images per Category



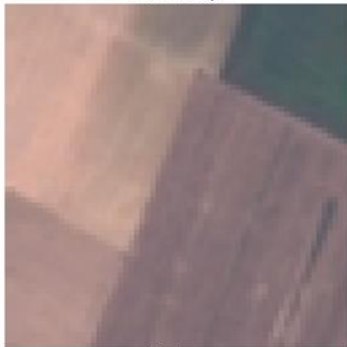
Category Distribution (%)



Exploratory Data Analysis Results

Sample Images From Each Category

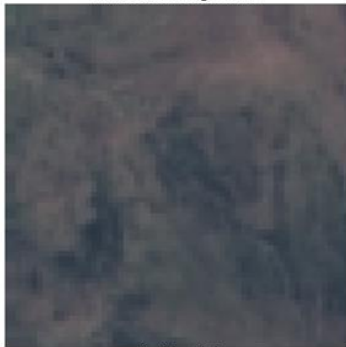
AnnualCrop



Forest



HerbaceousVegetation



Highway



Industrial



Pasture



PermanentCrop



Residential



River

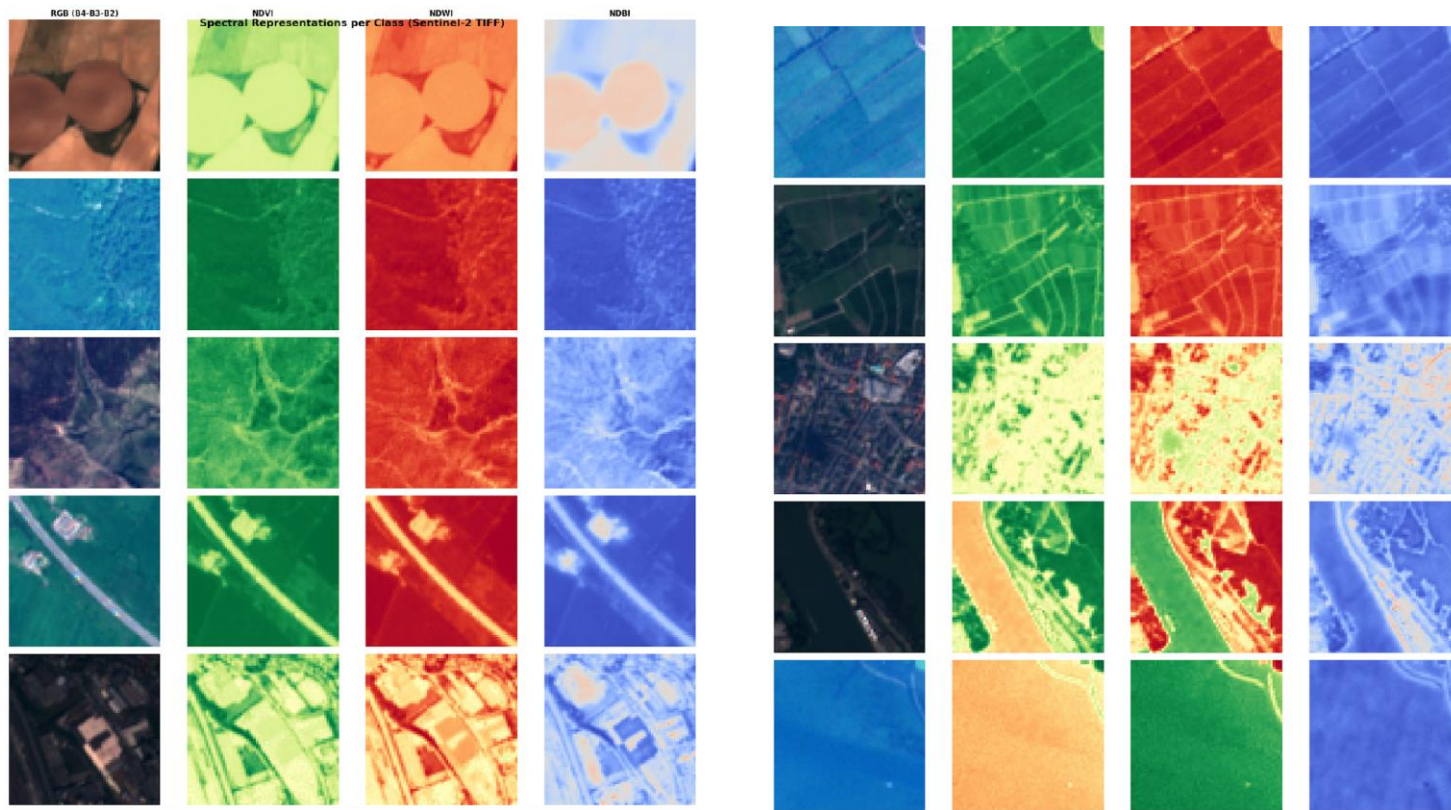


SeaLake



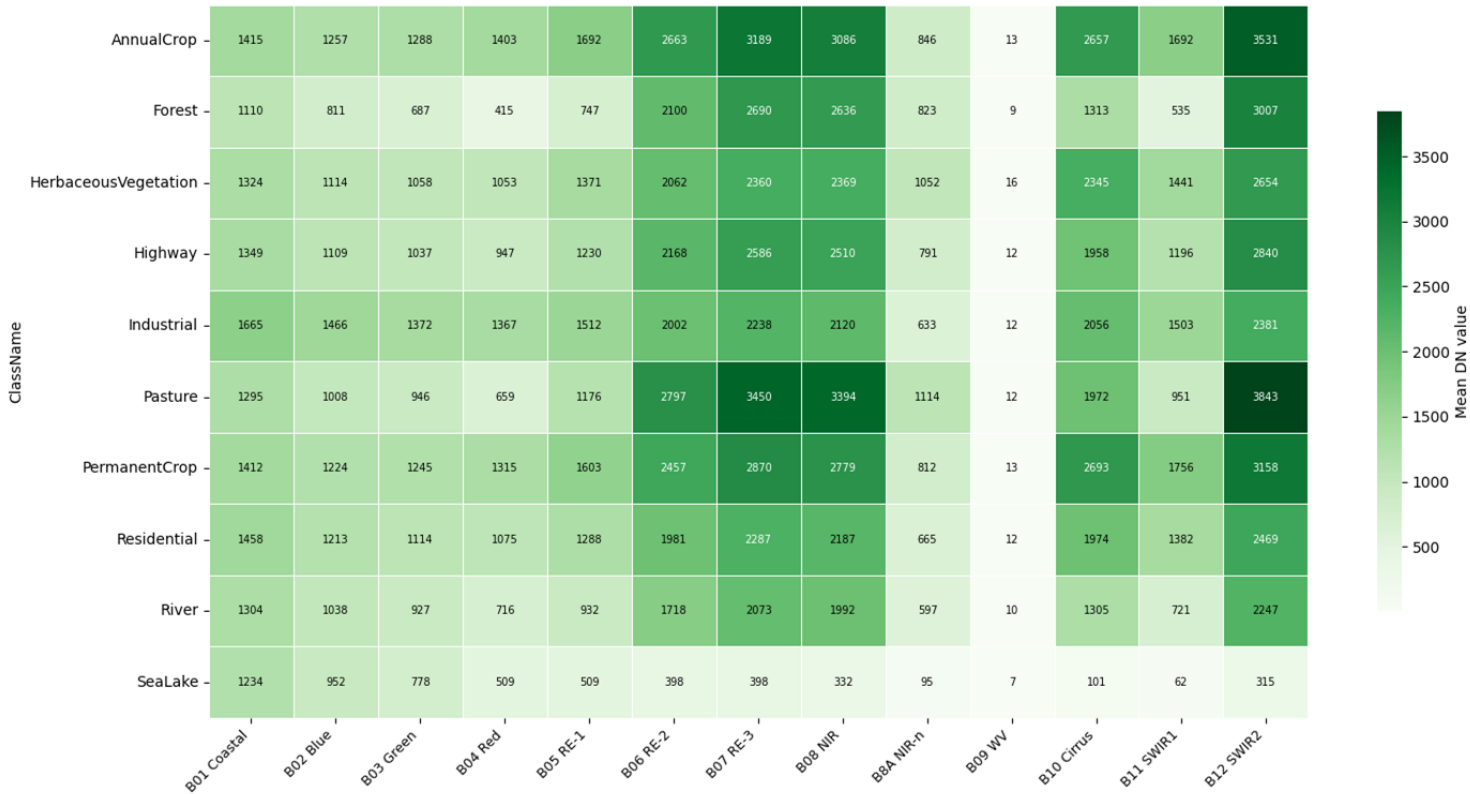
Exploratory Data Analysis Results

Displays RGB composite, NDVI, NDWI, and NDBI for one sample per class



Exploratory Data Analysis Results

Mean Spectral Band Values per Class



Exploratory Data Analysis Results

Top 5 Bands Per Class

Class Name	Rank 1	Rank 2	Rank 3	Rank 4	Rank 5
Annual Crop	B12 SWIR2	B07 RE-3	B08 NIR	B06 RE-2	B10 Cirrus
Forest	B12 SWIR2	B07 RE-3	B08 NIR	B06 RE-2	B10 Cirrus
HerbaceousVegetation	B12 SWIR2	B08 NIR	B07 RE-3	B10 Cirrus	B06 RE-2
Highway	B12 SWIR2	B07 RE-3	B08 NIR	B06 RE-2	B10 Cirrus
Industrial	B12 SWIR2	B07 RE-3	B08 NIR	B10 Cirrus	B06 RE-2
Pasture	B12 SWIR2	B07 RE-3	B08 NIR	B06 RE-2	B10 Cirrus
PermanentCrop	B12 SWIR2	B07 RE-3	B08 NIR	B10 Cirrus	B06 RE-2
Residential	B12 SWIR2	B07 RE-3	B08 NIR	B06 RE-2	B10 Cirrus
River	B12 SWIR2	B07 RE-3	B08 NIR	B06 RE-2	B10 Cirrus
Sea/Lake	B01 Coastal	B02 Blue	B03 Green	B04 Red	B05 RE-1

Model Development

Architecture: ResNet50 • Framework: PyTorch • Optimizer: Adam (lr=0.001) • Loss: CrossEntropyLoss • Epochs: 10

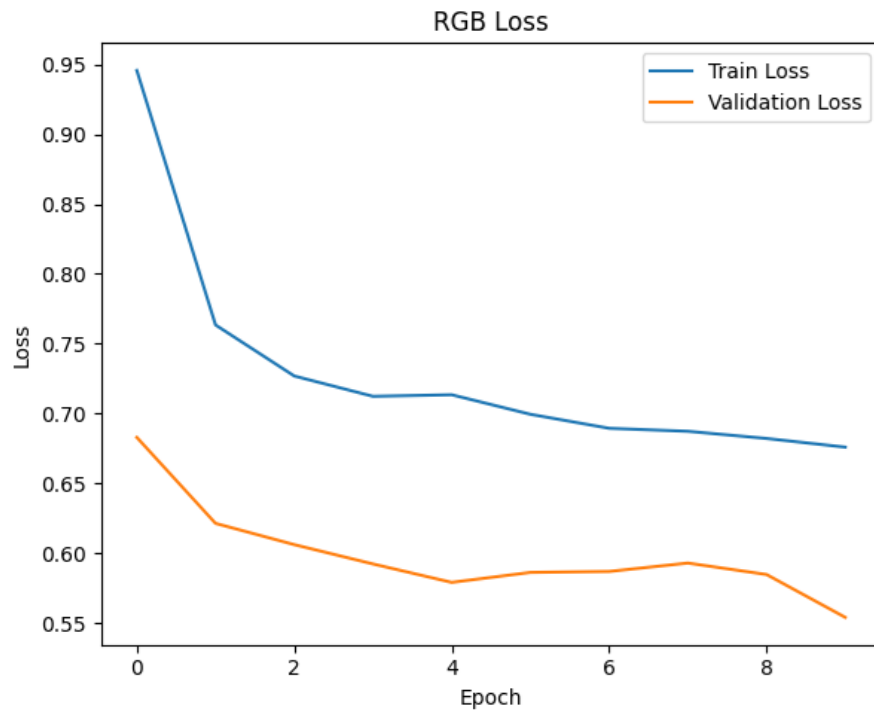
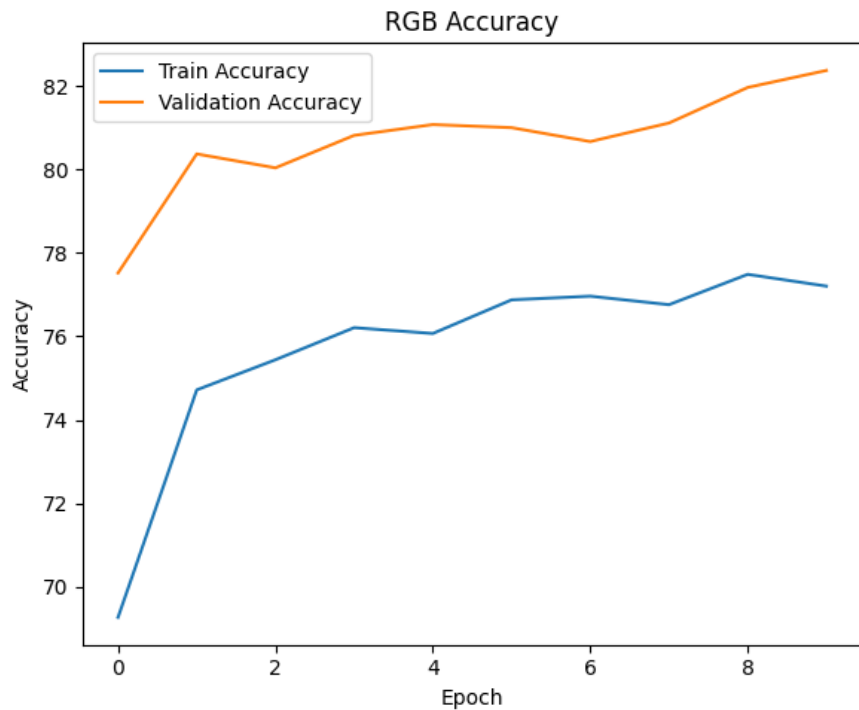
RGB Model		3-channel input
Base Model	ResNet50 (pretrained — ImageNet)	
Frozen Layers	All except FC head	
Custom Head	Linear(2048→512) → ReLU → Dropout(0.5) → Linear(512→N)	
Input	3-channel RGB images	
Trained Params	FC head only (fast fine-tuning)	

Multispectral Model		13-channel input
Base Model	ResNet50 (random weights — no pretrain)	
Modified Conv1	Conv2d(13→64, k=7, stride=2) — custom input	
Frozen Layers	All except conv1 + FC head	
Custom Head	Linear(2048→512) → ReLU → Dropout(0.5) → Linear(512→N)	
Trained Params	conv1 + FC head (13-band specialization)	

Training loop: forward pass → CrossEntropyLoss → backprop → Adam step | Tracked per epoch: train/val loss & accuracy | Saved to Google Drive (.pth)

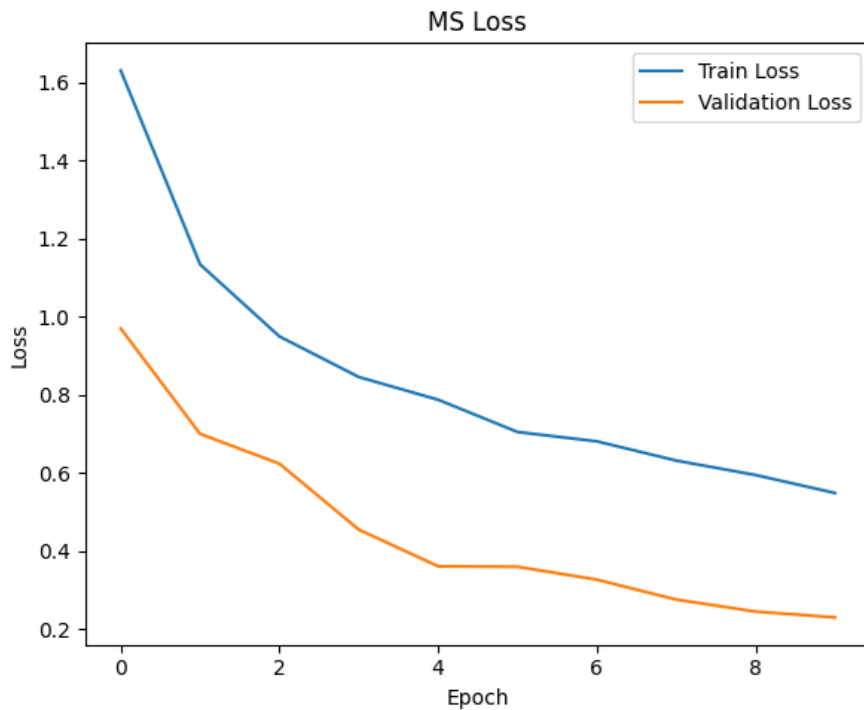
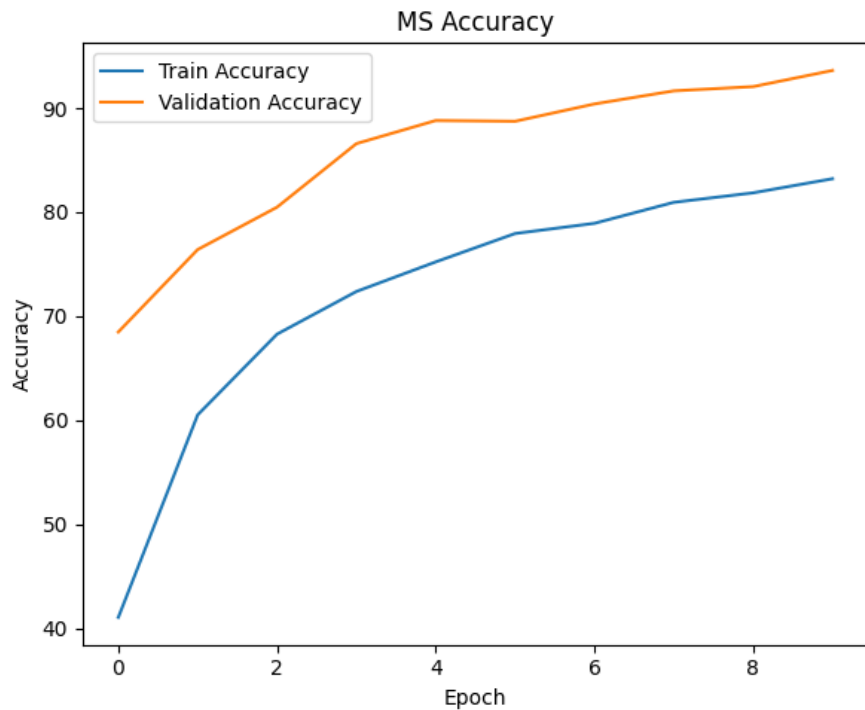
Training Results

RGB Model — Accuracy & Loss Curves



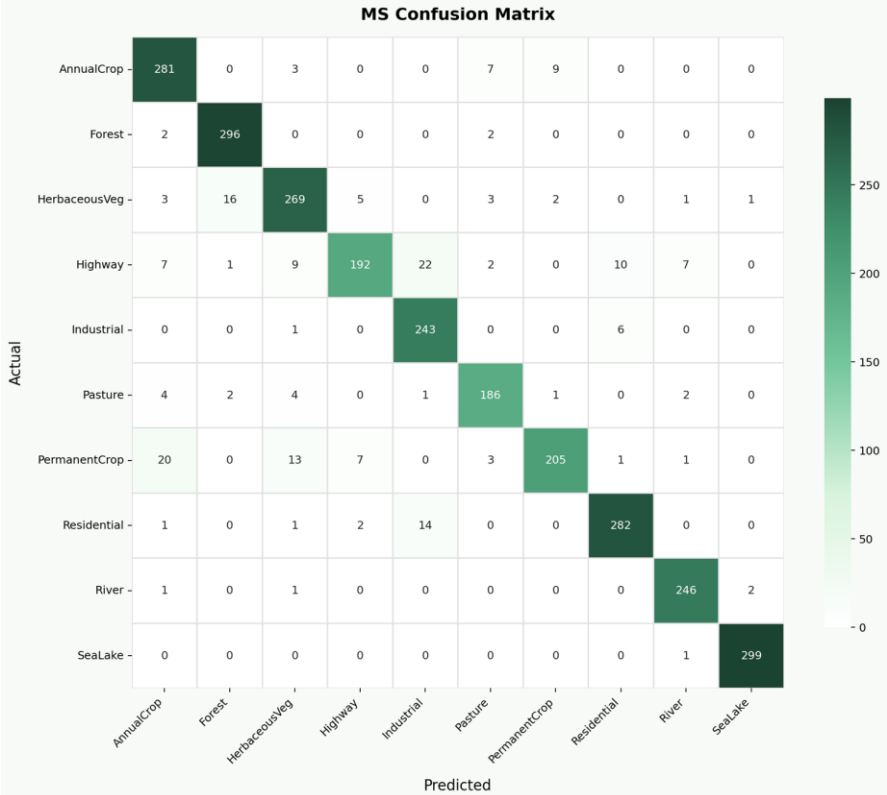
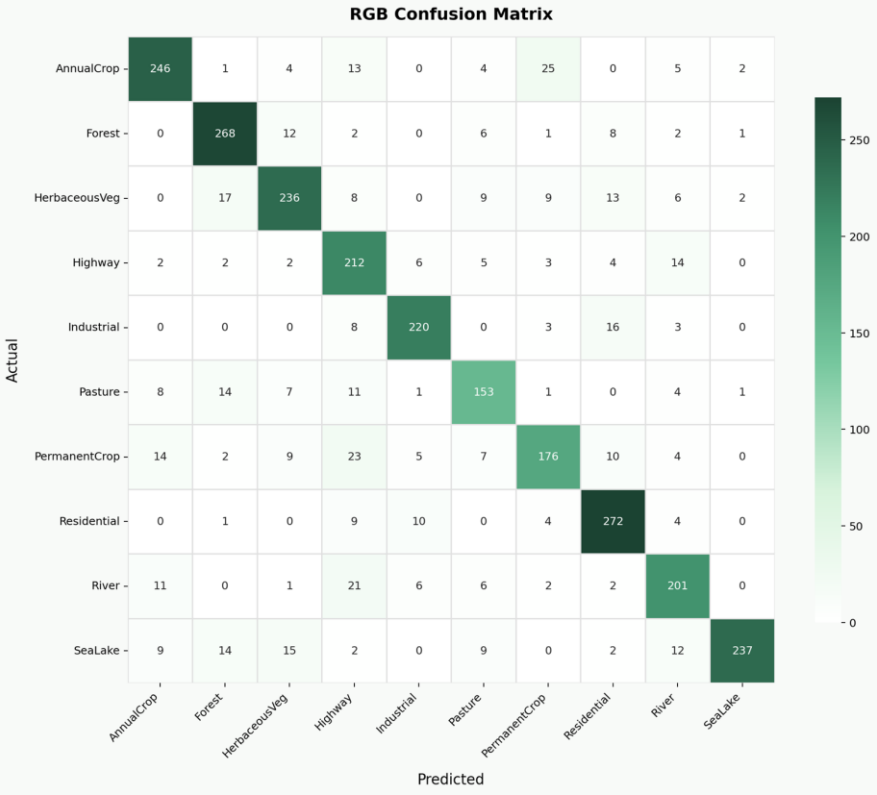
Training Results

MS Model — Accuracy & Loss Curves



Training Results

RGB & MS — Confusion Matrix



Model Results

Simple CNN

~84%

Accuracy

Baseline architecture
3 conv blocks
Overfit after ~10 epochs

CNN + Batch Norm

~56%*

Accuracy

Faster convergence
Still improving at 10 epochs
13h vs 34h training

★ BEST MODEL

Transfer Learning (ResNet-50)

~95%

Accuracy

Pre-trained on ImageNet
Fine-tuned on EuroSAT
Best generalization

** Batch Norm model still improving at epoch 10 — expected to exceed with more training*

Engineering the Pipeline

Development Environment & Project Structure



DEPI_DataScience_Project/

- **config.py** # Hyperparameters & paths
- **data_loader.py** # RGB & MS handling
- **model.py** # ResNet50 architecture
- **engine.py** # Training logic & AMP
- **train.py** # Command Line execution
- **app.py** # Streamlit Dashboard — classification & segmentation
- **requirements.txt** # Dependencies
- **README.md** # Project Documentation

Interactive Deployment: Streamlit Engine

Deployment & Interactive Dashboard

Streamlit Web Application

- **Dashboard Deployment:**
Deployed on Streamlit Cloud with a clean two-tab interface for instant access from any browser.
- **Real-Time Inference:**
Instant LULC classification across 10 land-cover categories for .jpg (RGB) or .tif (13-band MS) Sentinel-2 patches, with Top-3 probability scores.
- **Segmentation Map:**
Sliding-window analysis divides large images into 64x64 patches, classifies each independently, and renders a colour-coded land-cover map with area statistics.
- **Geospatial Insights:**
Toggle between RGB Natural Color and False Color Composite (NIR-Red-Green) views to highlight vegetation health across the uploaded scene.



Land Use & Land Cover Classification

Upload a satellite image and get instant classification with visual analysis.



Upload Image

Upload a RGB image



Upload

200MB per file • JPG, TIF



Upload an image to get started

Interactive Deployment: Streamlit Engine

Deployment & Interactive Dashboard



Oberdorla, Germany



<https://browser.data.space.copernicus.eu/>



Interactive Deployment: Streamlit Engine

Deployment & Interactive Dashboard

Single Prediction Segmentation Map



Classification Results



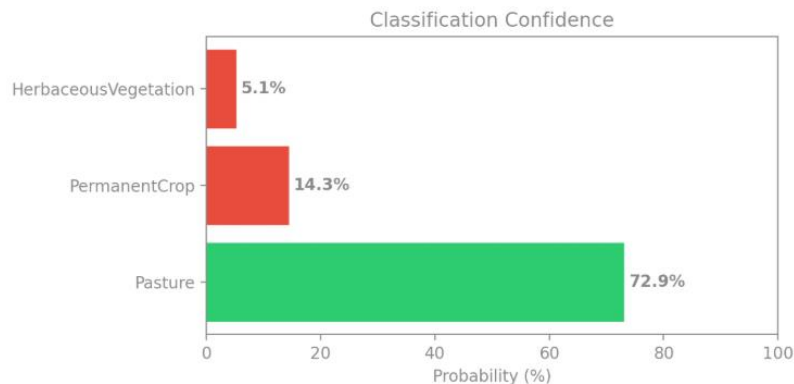
Satellite Image



Uploaded Patch



Top 3 Predictions



Interactive Dashboard UI Preview

Interactive Deployment: Streamlit Engine

Deployment & Interactive Dashboard



Detailed Analysis

Top Prediction

Pasture

↑ 72.9%

2nd Place

PermanentCrop

↑ 14.3%

3rd Place

HerbaceousVegetation

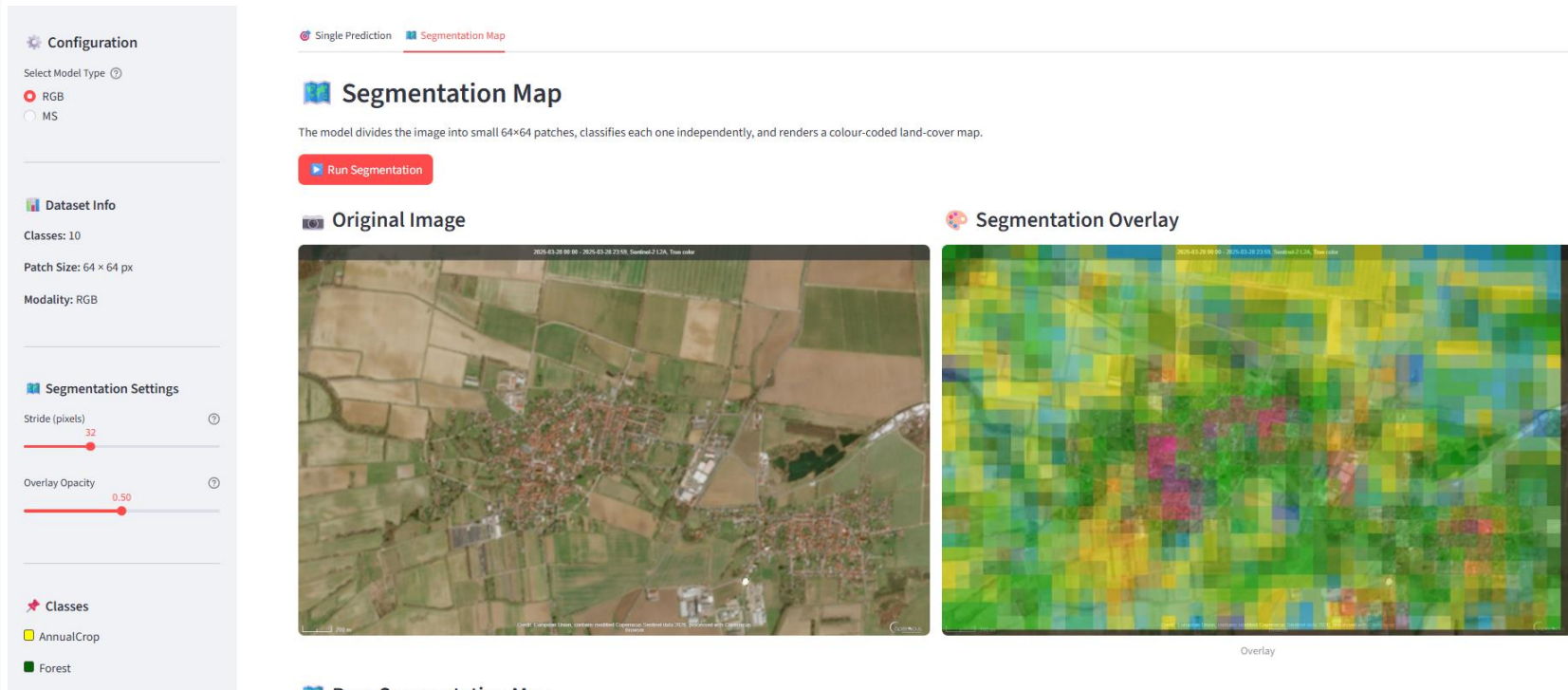
↑ 5.1%

All Class Probabilities

Class	Probability (%)
AnnualCrop	0.12%
Forest	1.21%
HerbaceousVegetation	5.08%
Highway	0.74%
Industrial	0.97%
Pasture	72.92%
PermanentCrop	14.29%
Residential	4.38%
River	0.25%
SeaLake	0.04%

Interactive Deployment: Streamlit Engine

Deployment & Interactive Dashboard



Interactive Dashboard UI Preview

Interactive Deployment: Streamlit Engine

Deployment & Interactive Dashboard

Configuration

Select Model Type ⓘ

☒ RGB
☐ MS

Dataset Info

Classes: 10

Patch Size: 64 × 64 px

Modality: RGB

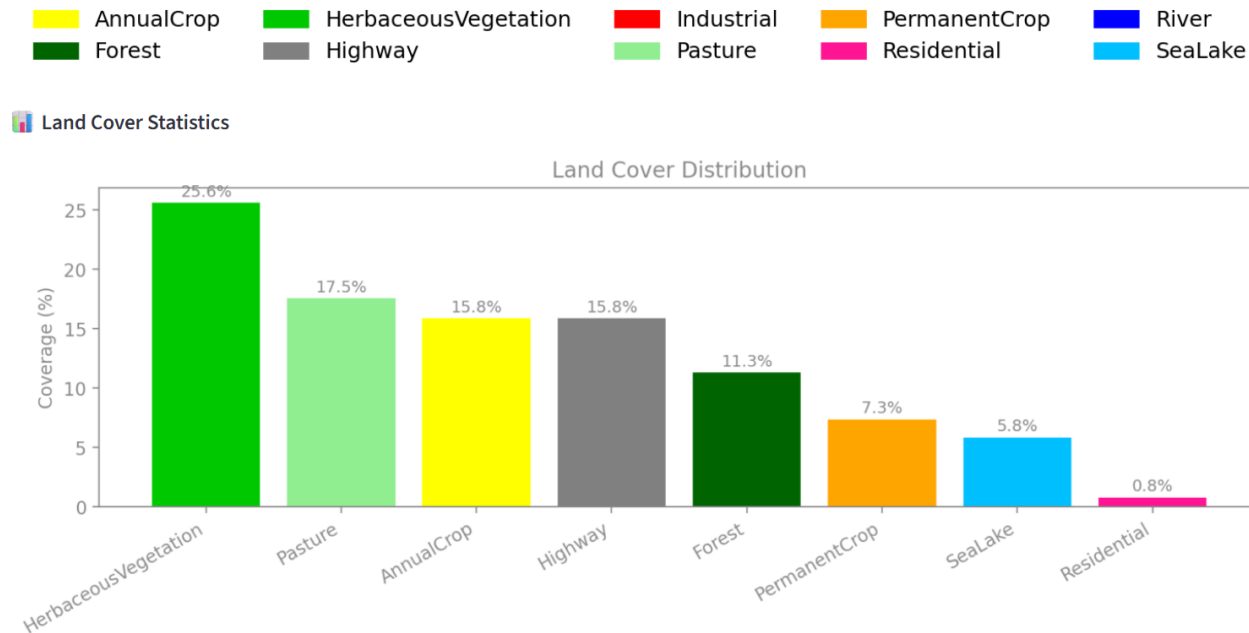
Segmentation Settings

Stride (pixels) ⓘ

Overlay Opacity ⓘ

Classes

- ☒ AnnualCrop
- ☒ Forest
- ☒ HerbaceousVegetation
- ☒ Highway

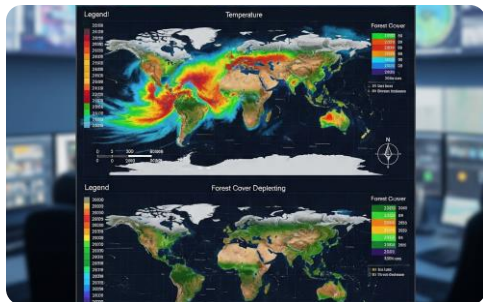


< Manage app

Interactive Dashboard UI Preview

Future Scope: Scaling the Geospatial Impact

Long-term Evolution & Regional Integration



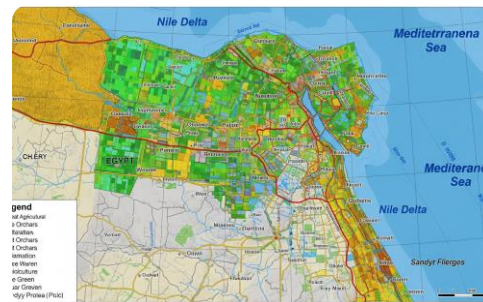
Climate & Change

Correlate LULC shifts with environmental variables for long-term predictive models and automated deforestation tracking.



Live Integration

Direct live connections with Sentinel-2 API and Google Earth Engine for on-demand regional analysis and cloud processing.



Regional Fine-Tuning

Adapting the ResNet50 backbone to Egyptian landscapes and Nile Delta topographies for agricultural monitoring and urban planning.

Real-World Impact: Transforming Geospatial Data into Action

Deploying Advanced ML for National Development & Sustainability



Combating Urban Sprawl

Automatically detecting illegal construction and urban encroachment on valuable agricultural land in the Nile Delta.



Precision Agriculture

Utilizing 13-band Multispectral and False Color Composites to monitor crop health and predict harvest yields.



Water Resource Management

Continuously tracking water levels and pollution in the Nile River and northern coastal lakes.



Smart City Planning

Assisting government entities in mapping terrain for new urban developments and optimizing highway routing.

THANK YOU!

For your time and attention

PROJECT TEAM

Nourhan Magdy

Abdelrahman Ashraf

Eslam Abdin

Heba Adel

Ahmed Abdelkhalek

Neamat Gamal



<https://github.com/nhahub/NHA-047-4>



<https://depidatascienceproject-zfm6ugw4g6h74ak3svcwfc.streamlit.app/>